

Cybersecurity (F26)

11th February 2026

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Course Plan

Lecture	Date	Teacher	Topic
01	2026-02-04	Sani	Introduction & Lab Set-up
02	2026-02-11	Sani	Vulnerability Assessment
03	2026-02-18	Sani	Penetration Testing
04	2026-02-25	Hiraku	Firewalls and IDS
05	2026-03-04	Hiraku	Malware & SQL
06	2026-03-11	Hiraku	Denial of Service
07	2026-03-18	Sani	Social Engineering, Phishing
08	2026-03-25	Sani	DDoS, MITM
09	2026-04-08	Hiraku	Cryptography
10	2026-04-15	Hiraku	Multi-party Computation
11	2026-04-22	Hiraku	Privacy-preserving Data Analysis
12	2026-04-29	Sani	Theat Modelling

Cybersecurity Incident of the Week

→ Replaced with Cyber Kill Chain analysis

→ **Period:** Just for this week

Vulnerability Assessment

Learning Objectives

By the end of this lecture, you should be able to:

- Understand vulnerability assessment fundamentals, and how it differs from penetration testing.
- Perform information gathering (reconnaissance) using both passive and active scanning.
- Conduct network scanning and enumeration using Nmap.
- Perform vulnerability scanning using OpenVAS/GVM and Nessus.
- Analyze and validate vulnerabilities and report findings.

Assessments

Penetration testing

- Test the security of the running system
- Authorized “simulated attack”

Is not a guarantee for security, but it should complement any security strategy, e.g., ongoing Threat Modelling as part of the development/deployment process.

Assessments

Penetration testing activities

- **Getting permission** - Maybe as part of a job?
- **Gathering Information**, e.g., about the company, ip addresses, web sites, etc., scan ip addresses, enumerating hosts & services, fingerprinting services, etc.
- **Threat Modelling**, e.g., what does the penetration test cover?
- **Vulnerability Analysis**, e.g., connecting the collected intel to possible vulnerabilities.
- **Exploitation**, e.g., carefully creating a proof of concept, demonstrating security problems without harming the target.
- **Reporting**, e.g., report and/or fix the problems

Assessments

Penetration testing activities

- Getting permission
- **Gathering Information**
 - ✓ General information
 - ✓ Scanning
 - ✓ Fingerprinting services
- **Threat Modelling**
- **Vulnerability Analysis**
- Exploitation
- Reporting

Vulnerability Assessment: involves discovering host and services, while investigating possible vulnerabilities.

Vulnerability Assessment

Gathering Information

Collect

- IP addresses
- Personal information
- Further information in registries (DNS, whois)
- Any other vital information that will aid your quest.

Vulnerability Assessment

Investigate

- Which hosts are up?
- Which ports are open?
- Which services are running?
- Service versions and operating systems?
- Network structure
- Further information on the public domain

Vulnerability Assessment

Scanning: Active vs. Passive Scanning

Passive Scanning

Gathering information about the target without actually scanning its network/system.

- Volunteered intel
- General & for social engineering
- Websites & online resources
 - ✓ Discussion boards
 - ✓ Chat rooms
 - ✓ archive.org
- Consistent job ads, e.g.,
 - ✓ New staff unfamiliar with organization or system
 - ✓ Skills, e.g., languages and infrastructure

Active Scanning

Actively scanning the target network/system using different tools.

- Extracting information not volunteered
- *Nmap*: information discovery
- *Nessus*: vulnerability assessment
- *Burp Suit & OWASP ZAP*: web scanners

Vulnerability Assessment

Identify Vulnerable Services

- Are there security concerns with the services?
- Not every open port is a security risk. E.g., Port 22 open on a managed server for SSH access
- It becomes a risk when service is vulnerable, misconfigured, or exposed to untrusted network.
- Needs research on the risks of,
 - ✓ e.g., certain versions of the system, services, etc ...
- Online resources provide plenty of information,
 - ✓ e.g., CVE, exploit-db, ...
- Tools can speed up the process dramatically,
 - ✓ e.g., Nessus, GVM, Legion, nmap, ...

Passive Scanning

Inspecting Websites



Platform Solutions Why Netcraft Resources Company

GET DEMO

Research Tools

Use our tools to find out what infrastructure and technologies any site is using, which sites are the most popular, and how to stay safe on the internet.



THREAT MAP

View Netcraft's detection and countermeasures in real time using our 3D visualization tool.

[→](#)



SITE REPORT

Using results from our internet data mining, find out the technologies and infrastructure of any site.

[→](#)



SEARCH DNS

Explore hostnames visited by users of the Netcraft extensions. Search by domain or keyword.

[→](#)



MOST POPULAR SITES

Find out which sites are most visited globally as determined by users of the Netcraft Extensions.

[→](#)



CYBERCRIME TRENDS

Use real-time data to discover where you are most likely to find cybercriminal activity.

[→](#)

SDU

netcraft.com

sdu.dk

#sdudk

Inspecting Websites

Site report for http://sdu.dk

Background			
Site title	Not Present	Date first seen	April 1998
Site rank	1288723	Primary language	English
Description	Not Present		
Network			
Site	http://sdu.dk	Domain	sdu.dk
Netblock Owner	Microsoft Corporation	Nameserver	ns1.sdu.dk
Hosting company	Microsoft - Europe West (Netherlands) datacenter	Domain registrar	dk-hostmaster.dk
Hosting country	nl	Nameserver organisation	whois.dk-hostmaster.dk
IPv4 address	52.233.201.66 (VirusTotal)	Organisation	Campusvej 55, Odense M, 5230, DK
IPv4 autonomous systems	AS8075	DNS admin	dns@sdu.dk
IPv6 address	Not Present	Top Level Domain	Denmark (.dk)
IPv6 autonomous systems	Not Present	DNS Security Extensions	Enabled
Reverse DNS	Unknown		

Inspecting Websites

Achives over 1 trillion websites

 INTERNET
ARCHIVE

 WEB

 TEXTS

 VIDEO

 AUDIO

 SOFTWARE

 IMAGES

 SIGN UP | LOG IN

 UPLOAD

ABOUTBLOGEVENTSPROJECTSHelpDONATECONTACTJOBSVOLUNTEER

Search the history of more than 1 trillion web pages on the Internet.

WayBackMachine

Enter URL or keywords



Celebrate 1 trillion
web pages archived!

Internet Archive is a non-profit library of millions of free texts, movies, software, music, websites, and more.

1T

48M

15M

13M

3.2M

1.3M

5.3M

283K

3.2M

Search

GO

Advanced Search

Archive News

Top News Stories About the Internet Archive: 2025

The Joyful Chaos of the Early Web: A Conversation with Creator Audrey Witters

DWeb in 2025: Looking Back at a Year of Decentralization

More posts

Inspecting Websites

Vital stats and info about the website

MIME-types

Any MIME-types

Year Start

1999

Year End

2025

Alltextimageapplicationvideoaudiochemical

Summary on MIME-types Count

Quick search on MIME-types...

Captures

Unique URLs

text/html

909,850

242,798

image/jpeg

88,179

30,552

image/png

40,233

9,378

application/pdf

32,702

18,077

application/octet-stream

14,696

73

image/gif

14,307

3,833

image/svg+xml

14,230

551

application/vnd.ms-fontobject

10,887

80

text/css

7,608

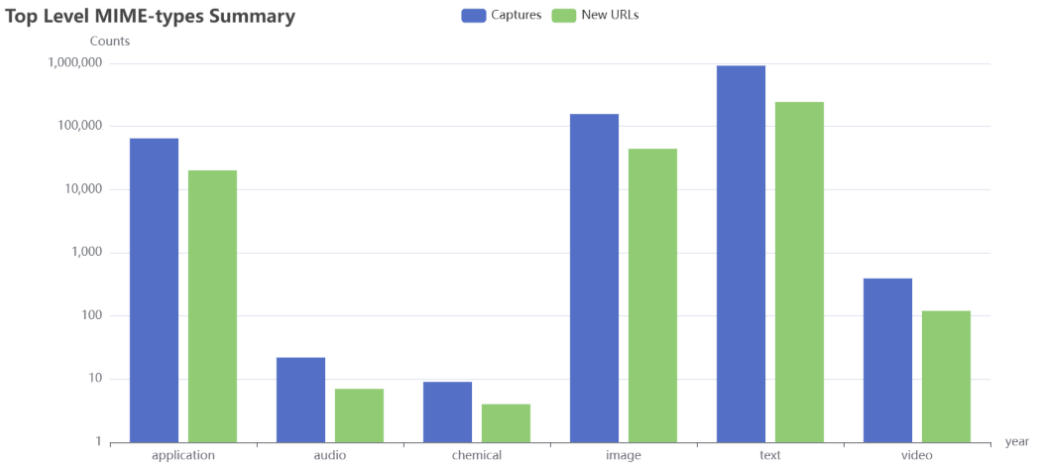
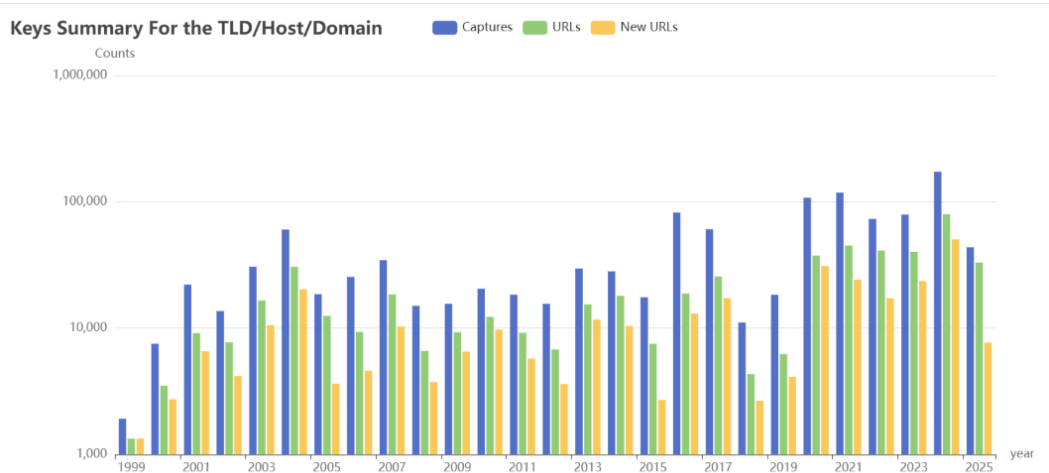
1,341

application/javascript

4,126

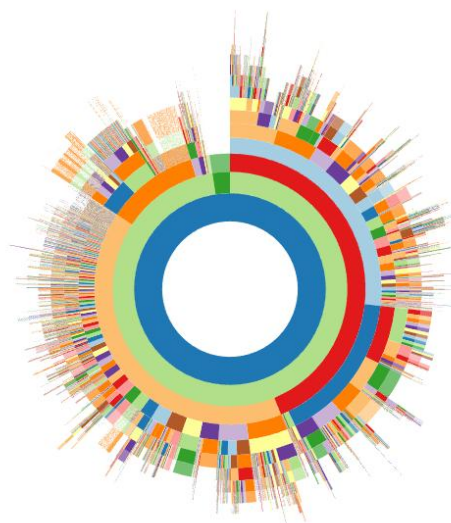
312

Captures



SDL

URLs

[illegible]

Search for Vulnerabilities

 SHODAN

Explore

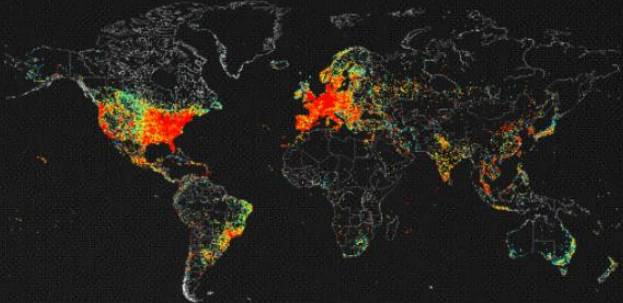
Pricing 

Type / to search 

Search Engine for the Internet of Everything

Shodan is the world's first search engine for Internet-connected devices. Discover how Internet intelligence can help you make better decisions.

SIGN UP NOW



// EXPLORE THE PLATFORM

 **Beyond the Web**

Websites are just one part of the Internet. Use Shodan to discover everything from power plants, mobile phones, refrigerators and Minecraft servers.

 **Monitor Network Exposure**

Keep track of all your devices that are directly accessible from the Internet. Shodan provides a comprehensive view of all exposed services to help you stay secure.

 **Internet Intelligence**

Learn more about who is using various products and how they're changing over time. Shodan gives you a data-driven view of the technology that powers the Internet.

Search for Vulnerabilities

 SHODAN

Explore

Downloads

Pricing [↗](#)

port:21 country:dk



Account

TOTAL RESULTS

13,033

TOP CITIES

Copenhagen	9,274
Århus	694
Randers	224
Aabenraa	193
Horsens	177

[More...](#)

TOP ORGANIZATIONS

Customer VPS	1,198
Hostnet B.V.	967
Hostnet BV Network Operations	804
Webdock.io ApS	708
Webdock AS44803	619

 View Report

 View on Map

 Advanced Search

Access Granted: Want to get more out of your existing Shodan account? Check out [everything you have access to.](#)

192.165.250.151

dc1sweb01.hexihost.dk
[Hexi Group ApS](#)

 Denmark, Kolding

 SSL Certificate

Issued By:
|- Common Name:
dc1sweb01.hexihost.dk

|- Organization:
None

Issued To:
|- Common Name:
dc1sweb01.hexihost.dk

|- Organization:
None

220----- Welcome to Pure-FTPd [privsep] [TLS] -----
220-You are user number 1 of 50 allowed.
220-Local time is now 18:50. Server port: 21.
220-This is a private system - No anonymous login
220-IPv6 connections are also welcome on this server.
220 You will be disconnected after 15 minute...

2026-02-06T17:50:51.323253

121.127.45.133

meltown.dathost.net
dathost.net
[Datacamp Limited](#)

 Denmark, Copenhagen



 SSL Certificate

Issued By:
|- Common Name:
E8

|- Organization:
Let's Encrypt

Issued To:

220 pyftplib 2.1.0 ready.
530 Anonymous access not allowed.
214-The following commands are recognized:
ABOR ALLO APPE AUTH CDUP CWD DELE EPRT
EPSV FEAT HELP LIST MDTM MFMT MKD MLSD
MLST MODE NLST NOOP OPTS PASS PASV PBSZ
PORT PROT PWD ...

2026-02-06T17:43:32.368543

Active Scanning

Active Scanning: Querying Databases

Domain Name System and IP Address Lookup

- Domain registers keep records of the domain they host.
 - ✓ E.g., information about the owner of the domain, contact information, etc.
- If it is a private information, you will see “Domains By Proxy”. Otherwise, you may be able to capture registered information of the owner of the domain.
- Usage at the command line:
 - `whois example.com`
 - `whois 192.168.0.1`
- **host & nslookup** can be used to query the Domain Name System (DNS) to obtain domain name or IP address mapping.

Active Scanning: Scanning Networks

NetDiscover

- Find hosts (i.e., IP's and MACs) on an internal network
- For example, find all the live hosts with IP addresses between 192.168.1.0 to 192.168.1.255 in a network.
 - ✓ `netdiscover -r 192.168.1.0/24 -f`
 - ✓ `-r`: specify which network to scan every IP for.
 - ✓ `-f`: tells NetDiscover not to try every IP on the network but just a few.
- **Be warned: this tool can cause a DoS on the target network.**

Scanning Networks

EtherApe

- Packet sniffer/network traffic monitoring tool
- Can replay from file
- Graphically displaying traffic
- Color coded protocols display
- Link layer, IP and TCP modes
- Supports Ethernet, FDDI, Token Ring, ISDS, PPP, SLIP, and WLAN devices.
- Support filtering

Scanning Hosts

Nmap: the network mapper

- Free and open-source tool for network discovery, network inventory, and security auditing.
- Can use raw packets.
- For finding hosts, services, operating systems, etc.
- Supports plugins
- Can also be used to assess vulnerabilities.

<https://nmap.org/>

Scanning Hosts

Nmap: Main Options

Host discovery

- -Px
- Many different types
- Stealthy to thorough
- Probe / ping e.g., ICMP
- -Pn-omit host discovery

Host scan

- -sY
- Find open ports
- Stealthy to thorough, e.g.,
 - ✓ Connect, Syn, Fin
- -sn-omit port scan

Scanning Hosts

Nmap: Scan Types (ICMP Scan)

- The ping scan sends a single ICMP echo request from the source to the destination device.
- A response from an active device returns an ICMP echo reply, unless the IP address is not available on the network or the ICMP protocol is filtered.
- `nmap -P <IP>` - regular scan
 - ✓ Receives a response? Device is online → standard port scan (TCP SYN if possible, else TCP connect)
 - ✓ Firewalls can block pings to prevent host discovery
 - ✓ Receives an ICMP Port Unreachable message? Port is closed
 - ✓ No response? Then either port is open but does not respond to Nmap or traffic is filtered.

Scanning Hosts

Nmap: Scan Types (Connect Scan)

- Fully connect to the target ip address and port
- Does a complete TCP handshake
- Most reliable, but can absolutely be detected.
- `nmap -sT <IP or IP range>` or
- `nmap -sV <IP or IP range>` to directly do a version scan
 - ✓ This determine the software versions on the target machines

Scanning Hosts

Nmap: Scan Types (Syn Scan)

- Sends SYN requests, but dose not complete handshake.
- Uses SYN-ACK requests to determine port state
- Connection not established $\Rightarrow$ application not alarmed

$\Rightarrow$ **Stealthier**

- `nmap -sS <IP or IP range>`
 - ✓ IDS or firewall logging can reveal this technique

Scanning Hosts

Nmap: Scan Types (Fin Scan)

- Sends a FIN packet to target [[nmap docs](#)]
- Request for closing an **open** connection
- If port is not listening, returns RST (reset). Otherwise, packet is dropped.

⇒ **Even more stealthier**

- `nmap -sF <IP or IP range>`
 - ✓ IDS or firewalls can also reveal this technique

Scanning Hosts

Nmap: Other Port Scanning Options

- Default: “the most common 1000 ports for each protocol”
- Specify individual ports with `-p<port>`
- Specify range with `-p<start>-<end>`
- Scan all ports with `-p-`
- Explicitly scan “port” 0
- `nmap -sS <IP or IP range> -oA <filename>`
 - ✓ `-sS`: TCP SYN scan
 - ✓ `-oA`: log results in all three formats – normal `.nmap`, XML, grepable `.gnmap`

Scanning Hosts

Nmap: Scripts

- Extended functionality via scripted modules
- List of nmap scripts:
 - [nmap book: list with descriptions](#)
 - locally: `ls /usr/share/nmap/scripts`
- List can be overwhelming
- Help per script: `--script-help <script name>`
 - ✓ E.g., `nmap -script smb-enum-users.nse -p 445 <target ip>`

Scanning Hosts

Nmap: Other Scan Options

- Ping: `nmap -sn <IP>` # Host discovery, no port scans
- UDP scan: `nmap -sU <IP>` # Connectionless, with no guaranteed outcome
- More information with `-h` or `-help`, or `man nmap`
- Nmap [Cheat Sheet](#) (unofficial)
- Many more to be found online:
 - ✓ [Nmap book](#)
 - ✓ [Nmap official website](#)

Scanning Hosts

Nmap: Other Scan Options (Not comprehensive)

- -O #detect operating system
- -sP #'ping host', host discovery
- -sT #TCP connect scan
- -sS #SYN scan
- -sF #FIN scan
- -sX #xmas scan
- -sN #Null scan
- -sU #UDP scan
- -sO #Protocol scan
- -sA #ACK scan
- -sW #TCP Window scan
- -sR #RPC scan
- -sL #list/DNS scan
- -sI #idle scan
- -Po #Don't ping
- -PT #TCP ping
- -PS #SYN ping
- -PI #ICMP ping
- -PB #TCP and ICMP ping
- -PM #ICMP netmask
- -oN #Normal output
- -oX #XML output
- -oG #grepable output
- -oA #all output formats
- -T #timing
 - T0 #paranoid
 - T1 #sneaking
 - T2 #polite
 - T3 #normal
 - T4 #aggressive
 - T5 #insane

Scanning Hosts

Nmap: The Scan Riddle

Question: What does this command do?

- `nmap -sS -sV -A -p- <IP> \ -v -oN scan-results.txt`
- **Hint:** Look up flags using `--help` or `man nmap`

Tools

List provided by OWASP

OWASP provides a comprehensive list of commercial and open-source tools with varying capabilities

Vulnerability Scanning Tools

OWASP

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Vulnerability Scanning Tools

Description

Web Application Vulnerability Scanners are automated tools that scan web applications, normally from the outside, to look for security vulnerabilities such as Cross-site scripting, SQL Injection, Command Injection, Path Traversal and insecure server configuration. This category of tools is frequently referred to as [Dynamic Application Security Testing \(DAST\)](#) Tools. A large number of both commercial and open source tools of this type are available and all of these tools have their own strengths and weaknesses. If you are interested in the effectiveness of DAST tools, check out the OWASP Benchmark project, which is scientifically measuring the effectiveness of all types of vulnerability detection tools, including DAST.

Here we provide a list of vulnerability scanning tools currently available in the market.

Disclaimer: The tools listing in the table below are presented in alphabetical order. **OWASP does not endorse any of the Vendors or Scanning Tools by listing them in the table below.**

OWASP is aware of the [Web Application Vulnerability Scanner Evaluation Project \(WAVSEP\)](#). WAVSEP is completely unrelated to OWASP and we do not endorse its results, nor any of the DAST tools it evaluates. However, the results provided by WAVSEP may be helpful to someone interested in researching or selecting free and/or commercial DAST tools for their projects. This project has far more detail on DAST tools and their features than this OWASP DAST page.

Tools Listing

Name/Link	Owner	License	Platforms	Note
Acunetix	Acunetix	Commercial	Windows, Linux, MacOS	Free (Limited Capability)
Aikido DAST	Aikido Security	Commercial or Free	SaaS, On-Premises, IDE Plugin	Aikido Security is a developer-friendly software security platform that combines multiple different scanners to deliver real-time alerts in a central system
Akto	Akto	Commercial	SaaS or On-Premises	500+ tests, 2 weeks free trial available

The OWASP® Foundation works to improve the security of software through its community-led open source software projects, hundreds of chapters worldwide, tens of thousands of members, and by hosting local and global conferences.

Upcoming OWASP Global Events

[OWASP Global AppSec EU 2026 - Vienna, Austria](#)

- June 22-26, 2026

[OWASP Global AppSec USA 2026 - San Francisco, CA](#)

- November 2-6, 2026

[OWASP Global AppSec EU 2027 - Vienna, Austria](#)

- June 21-25, 2027

[OWASP Global AppSec USA 2027 - Atlanta, GA](#)

- September 20-24, 2027

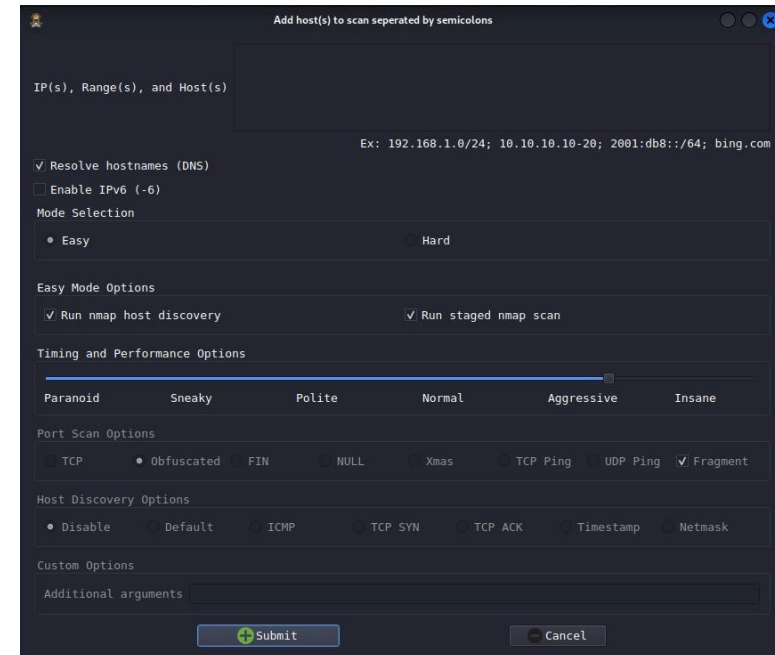
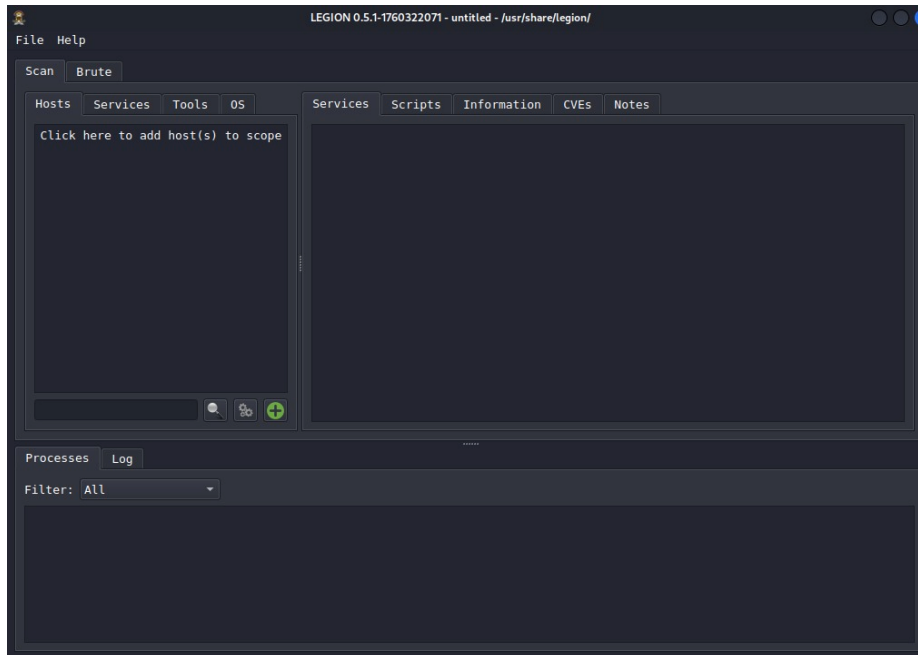
[OWASP Global AppSec EU 2028 - Vienna, Austria](#)

- June 19-23, 2028

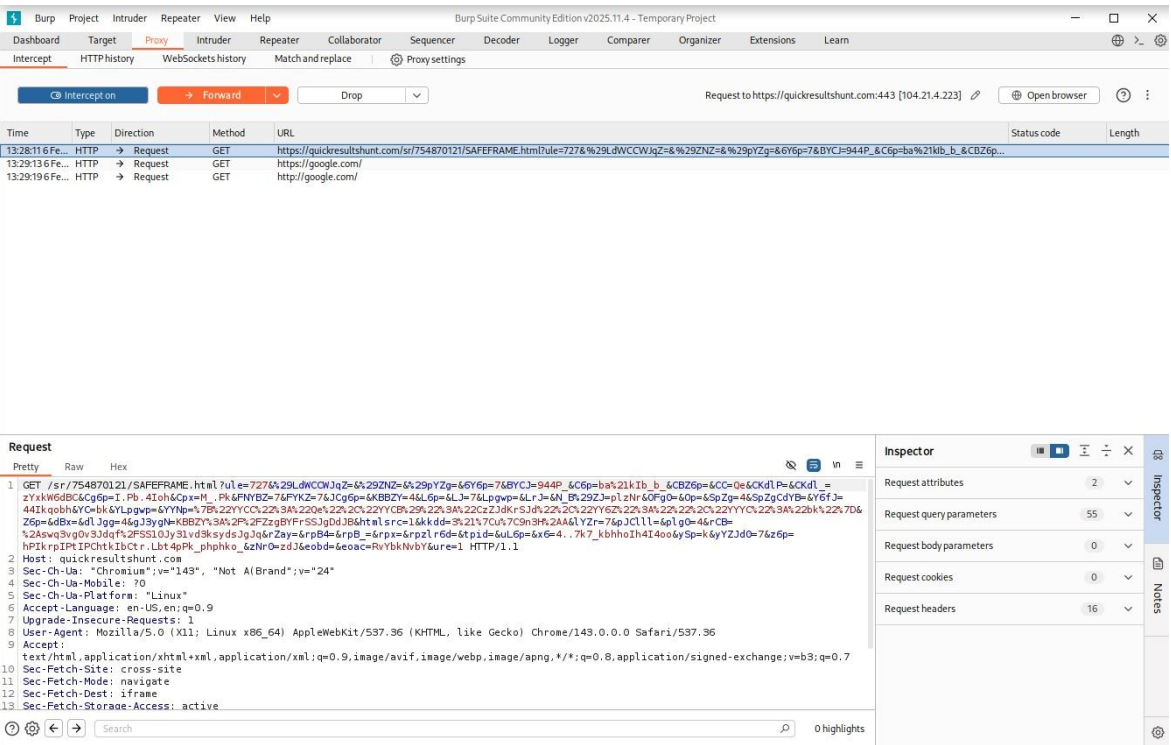
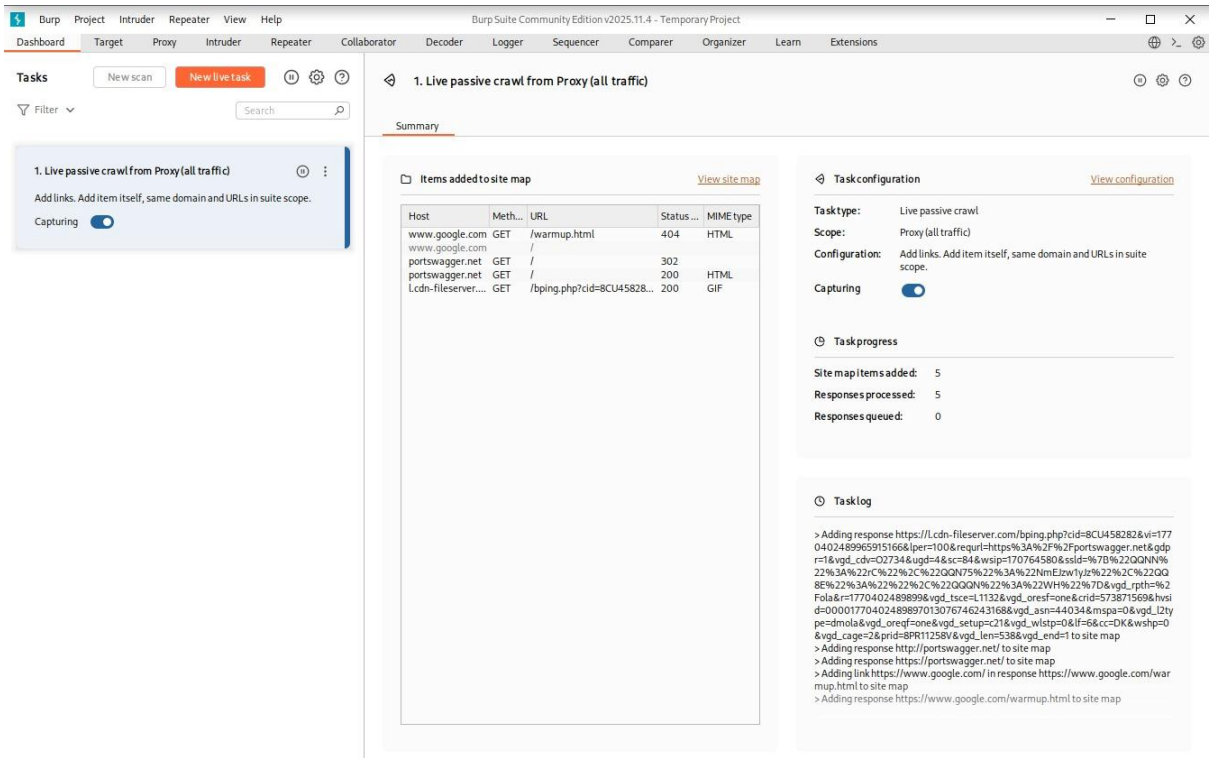
Tools

Legion

- Tool for automating assessments
- Include set of python scripts, UI, and incorporates alot of external tools.



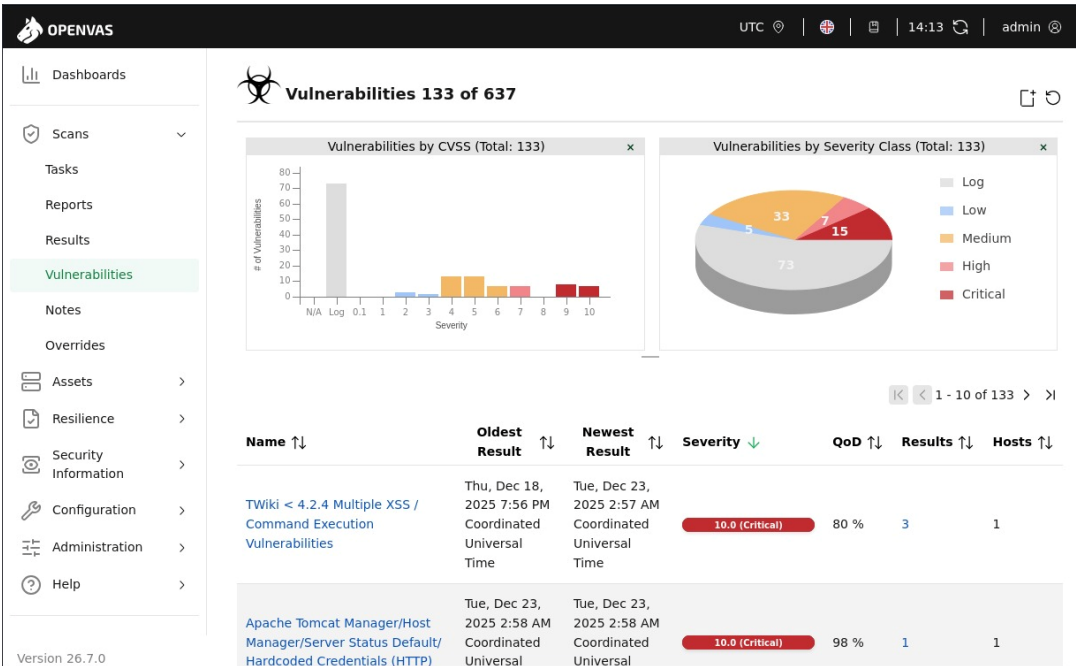
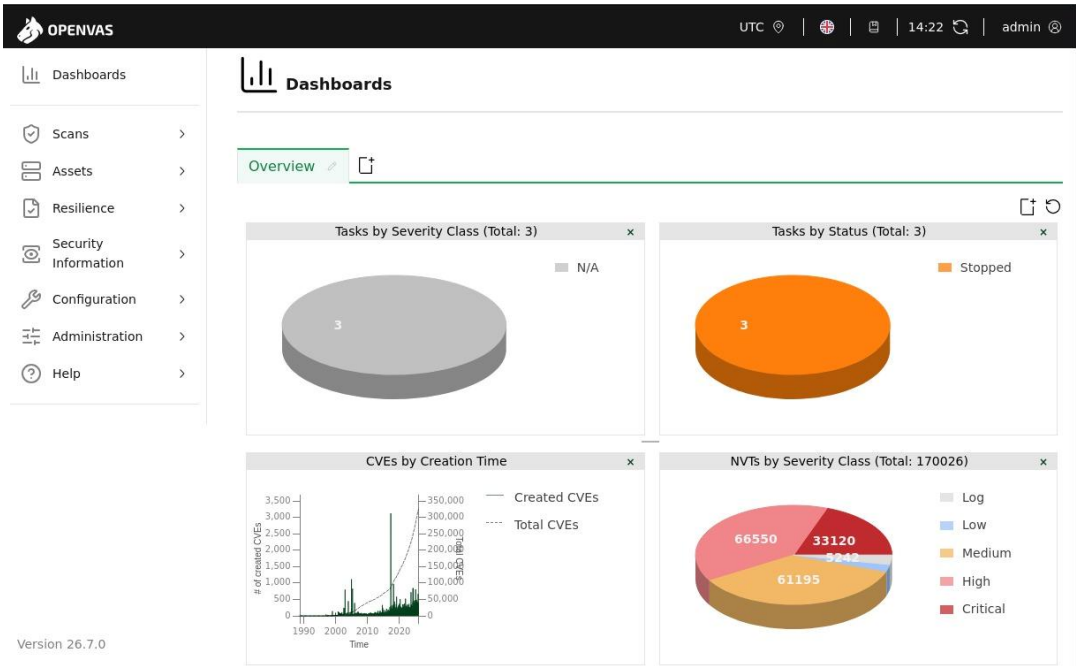
- Tool for web attack



Tools

OpenVAS by Greenbone (GVM)

- Tool for automating assessments
- Fork of Nessus, with a web interface.



Tools

Nessus

- Nessus provides description of the vulnerability and a solution
- It also supports authenticated scan. Thereby detecting more vulnerabilities

The image displays three screenshots of the Tenable Nessus Essentials Plus web interface, showing scan results for Metasploitable2 and a Lab Network.

Top Left Screenshot: Metasploitable2 Scan Results

- Hosts:** 172.30.1.21 (Auth: Fail, Vulnerabilities: 137)
- Scan Details:**
 - Policy: Basic Network Scan
 - Status: Completed
 - Severity Base: CVSS v3.0
 - Scanner: Local Scanner
 - Start: January 1 at 12:21 PM
 - End: January 1 at 12:37 PM
 - Elapsed: 16 minutes
- Vulnerabilities:** 69 (Critical: 16, High: 6, Medium: 22, Low: 8, Info: 137)

Top Right Screenshot: Metasploitable2 / Plugin #201352 Details

- Vulnerabilities:** 69
- Plugin Details:**
 - Severity: Critical
 - ID: 201352
 - Version: 1.2
 - Type: combined
 - Family: General
 - Published: July 9, 2024
 - Modified: March 26, 2025
- Description:** According to its version, Canonical Ubuntu Linux is 8.04.x. It is, therefore, no longer maintained by its vendor or provider. Lack of support implies that no new security patches for the product will be released by the vendor. As a result, it may contain security vulnerabilities.
- Solution:** Upgrade to a version of Canonical Ubuntu Linux that is currently supported.
- See Also:** <https://www.nessus.org/u?3b9b2d2e>
- Output:**

Bottom Screenshot: Lab Network Authenticated Vul Scan Results

- Hosts:** 172.30.1.22, 172.30.1.21, 172.30.1.20, 172.30.1.1 (Auth: Pass, Vulnerabilities: 129, 136, 130, 17, 14, 16, 6)
- Scan Details:**
 - Policy: Basic Network Scan
 - Status: Completed
 - Severity Base: CVSS v3.0
 - Scanner: Local Scanner
 - Start: Today at 5:23 AM
 - End: Today at 6:06 AM
 - Elapsed: 42 minutes
- Vulnerabilities:** 136 (Critical: 16, High: 6, Medium: 22, Low: 8, Info: 137)

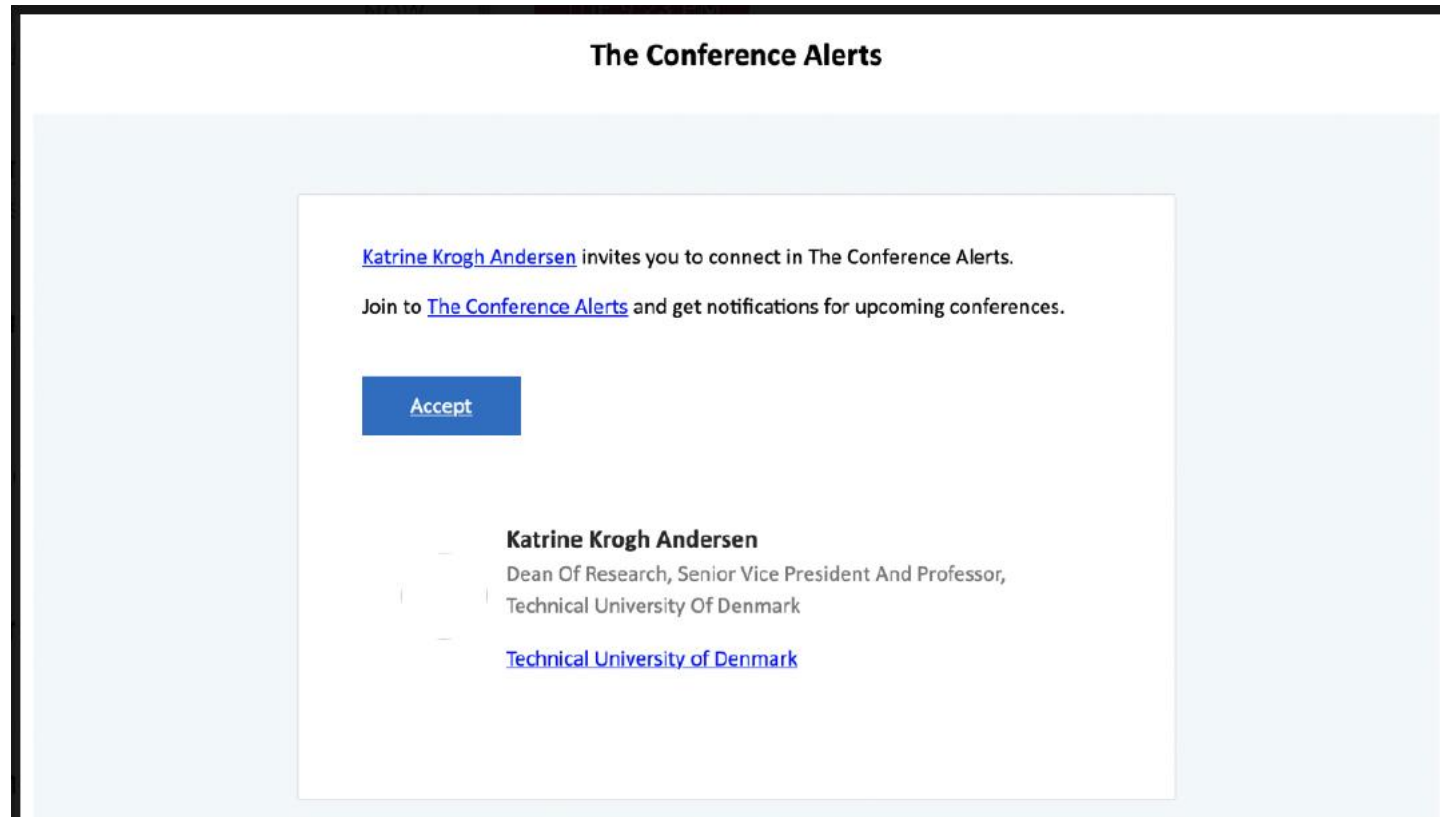
Phishing

Trick victims into giving out sensitive information, e.g., account information, bank details, etc.

- **Aim:** the attacker can then exploit the information gathered to masquerade as the victim by e.g., accessing social media accounts, online bank account, etc.
- **How:**
 - ✓ Fake website - that mimics a legitimate website, e.g., a seemingly official email from a bank that asks you to update your account information.
 - ✓ Trick you into downloading malware.
- **Spear phishing:** is phishing directed at a specific individual or company
- Not to be confused with Spam: which is unsolicited messages, e.g., email, texts, etc.

Phishing

Example: phishing email received at DTU



Metasploit Framework – Sneak Peak

- Metasploit is a framework that provides exploits for vulnerabilities.
- First released in 2003 (owned by Rapid7)

Modules:

- **Auxiliary:** Include port scanners, fuzzers, sniffers, spoofers, and more
- **Exploits:** are modules that use payloads
- **Payloads:** Consists of code that runs remotely

```
(kali㉿kali)-[/usr/share/metasploit-framework/modules]  
$ ls  
auxiliary  encoders  evasion  exploits  nops  payloads  post  README.md
```

Class Demo

Assessing Vulnerabilities

Thank You!